



Paper ID : 260688

Printed Page: 1

Subject Code:BP801T

Roll No:

BPHARM**(SEM. VIII) THEORY EXAMINATION 2025-26****BIOSTATISTICS AND RESEARCH METHODOLOGY****TIME: 3 HRS****M.MARKS: 75****Note:** 1. Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****10 x 2 = 20**

a.	What do you mean by 'frequency distribution'?
b.	Define Karl Pearson's coefficient of correlation.
c.	Cite an example of multiple regression.
d.	Differentiate between 'normal distribution' and 'binomial distribution'.
e.	State the applications of Mann-Whitney U test.
f.	Mention the need for design of experiments.
g.	Name two major statistical software used for clinical trial approach.
h.	What do you mean by hypothesis testing in multiple regression models?
i.	State the advantages of factorial design.
j.	Define 2^3 factorial design.

SECTION B**2. Attempt any two parts of the following:****2 x 10 = 20**

a.	The average weight of acetaminophen in a medication is supposed to be 80 mg, but the average weight of 50 random samples is 79.95 mg with a standard deviation of 0.18. i) Identify the null and alternative hypothesis. ii) Under what conditions is the null hypothesis accepted? iii) Determine if these differences in average weight are significant.
b.	Write a brief note on Hypothesis testings in multiple regression models.
c.	Describe in brief an optimization technique for pharmaceutical preparations.

SECTION C**3. Attempt any five parts of the following:****7 x 5 = 35**

a.	Explain the properties of Normal distribution and Poisson's distribution.
b.	Write in brief the significance of conducting one way and two ways ANOVA.
c.	Describe the significance of using plagiarism software in pharmaceutical research.
d.	Explain the steps involved in statistical analysis of data using SPSS technique.
e.	Describe the central composite design used in Response Surface methodology.
f.	Differentiate between response surface plot and counter plot graph.
g.	State and explain the steps involved in 2^2 factorial designs.